MSc Dissertation Topic List 2026/27

Please see below a recommended list of dissertation topics for 2026/27, and contact details for the respective supervisor. Please feel free to email them directly or book an in-person meeting with them within office hours.

You may also propose your own dissertation topic, provided you can find a supervisor. The list of topics below will give you an idea of staff research interests. For further information, please visit the academic staff webpages https://www.lse.ac.uk/Statistics/People.

Important: Your chosen supervisor must be a member of academic staff within the Department of Statistics.

Optimal Algorithms and Statistical Limits for Ranking from Pairwise Comparisons: Applications to Large Language Models

Supervisor: Dr Mona Azadkia, M.Azadkia@lse.ac.uk

Ranking from noisy and incomplete pairwise comparisons is a fundamental problem in statistics and machine learning, with applications ranging from recommendation systems and search engines to the evaluation and ranking of large language models (LLMs). In particular, modern LLM evaluation often relies on human preferences or comparisons generated by other LLMs (*LLM-as-a-judge*), raising important questions about the statistical accuracy, robustness, and computational efficiency of ranking algorithms.

In this dissertation, the candidate will:

1. Study the theoretical foundations of ranking from pairwise comparisons and preference lists, focusing on top-K identification and the statistical limits of ranking.
2. Implement and analyse spectral and maximum likelihood-based ranking methods, investigating their statistical properties, convergence behaviour, and sample complexity.
3. Compare these algorithms empirically under varying levels of data sparsity, noise, and model misspecification, with potential applications to LLM evaluation and ranking.
4. Explore extensions to modern preference-learning settings, such as LLM-generated comparisons, biased or inconsistent evaluators, and adaptive comparison selection, potentially developing methods with improved robustness or statistical efficiency.

The project combines theoretical analysis with computational experiments and offers opportunities to investigate contemporary challenges in LLM evaluation, preference learning, and reliable ranking.

Selected References:

Chen, Y., Fan, J., Ma, C., and Wang, K. (2019). *Spectral method and regularized MLE are both optimal for top-K ranking.* Annals of Statistics, 47(4), 2204–2235.

Chiang, W.-L., et al. (2024). *Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference.* ICML.

Shi, L., et al. (2025). *Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge.* IJCNLP-AACL.

Xu, M., Tan, X., Wu, J., and Zhou, D. (2026). *A Judge-Aware Ranking Framework for Evaluating Large Language Models without Ground Truth.* ICML.

Alimohammadi, Y., and Asgari, K. *Mallows Model with Learned Distance Metrics: Sampling and Maximum Likelihood Estimation.*

Kumar, R., and Vassilvitskii, S. *Generalized Distances between Rankings.*

Measuring and Testing Dependence Between Networks

Supervisor: Dr Mona Azadkia, M.Azadkia@lse.ac.uk

Networks arise naturally in many areas of statistics and data science, including social networks, neuroscience, finance, and biological systems. In many applications, several networks are observed over the same set of individuals or entities, and an important statistical question is whether and to what extent their connectivity structures are dependent. For example, one may wish to investigate whether patterns of friendship are associated with patterns of communication, or whether different brain connectivity networks exhibit related structures.

Measuring dependence between networks presents challenges that do not arise in classical multivariate settings. Network observations are typically high-dimensional, their edges may be dependent, and associations between networks can involve complex nonlinear relationships. Traditional correlation measures are therefore not always adequate for capturing the underlying dependence.

This dissertation will investigate statistical methods for measuring and testing dependence between networks, with particular emphasis on recent developments in nonparametric dependence measures and statistical network analysis. The candidate will study approaches based on graph correlation, distance covariance, and spectral graph embeddings, examining how existing dependence measures can be adapted to network-valued data. Depending on the scope of the project, the dissertation may also consider conditional dependence between networks, where the aim is to assess whether two networks remain associated after accounting for shared characteristics or network structure.

The project will combine a review of relevant statistical methodology with a simulation study comparing the performance of selected methods under different network models, such as Erdős–Rényi random graphs, stochastic block models, and random dot product graphs. Particular attention will be given to the ability of different methods to detect linear and nonlinear dependence, their statistical power, and their robustness to network sparsity and heterogeneity.

Ambitious candidates may investigate extensions of existing dependence measures to network settings, explore dependence in multilayer networks, or develop procedures for testing conditional independence between networks.

Reference reading:

Shen, C., Arroyo, J., Xiong, J., and Vogelstein, J. T. (2026). Community correlations and testing independence between binary graphs. Applied Network Science.

Pantazis, K., Athreya, A., Arroyo, J., Frost, W. N., Hill, E. S., and Lyzinski, V. (2022). The importance of being correlated: Implications of dependence in joint spectral inference across multiple networks. Journal of Machine Learning Research, 23, 1–77.

Lee, Y., Shen, C., Priebe, C. E., and Vogelstein, J. T. (2019). Network dependence testing via diffusion maps and distance-based correlations. Biometrika, 106(4), 857–873.

Azadkia, M. and Chatterjee, S. (2021). A simple measure of conditional dependence. The Annals of Statistics, 49(6), 3070–3102.

Chatterjee, S. (2021). A new coefficient of correlation. Journal of the American Statistical Association, 116(536), 2009–2022.

Székely, G. J., Rizzo, M. L., and Bakirov, N. K. (2007). Measuring and testing dependence by correlation of distances. The Annals of Statistics, 35(6), 2769–2794.

Brownian (and related?) Spiders

Supervisor: Dr Erik Baurdoux, e.j.baurdoux@lse.ac.uk

Various interesting properties have been derived, including its maximal range per root of the expected stopping time going back to Walsh and Dubins. Various directBrownian spiders can be considered as a Brownian motion running along n edges. Various interesting properties have been derived (e.g. Ernst 2016, Peskir 2025), including its maximal range per root of the expected stopping time going back to Walsh and Dubins (1978). Various directions can be considered in this dissertation, e.g. through simulation or possible extensions to include jumps in the process.ions can be considered in this dissertation, e.g. through simulation or possible extensions to include jumps in the process.

Prerequisite: ST409

References include:

Walsh, J.B. (1978). A diffusion with a discontinuous local time. Astérisque, 52 (37–45).

Ernst, P. (2016) Exercising control when confronted by a (Brownian) spider. Oper. Res. Lett. 44 (487–490).

Peskir: preprint

https://personalpages.manchester.ac.uk/staff/goran.peskir/dubins-walsh.pdf

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Asymptotic Optimality of Dynamic First-fit Packing on the Half-axis

Supervisor: Dr Erik Baurdoux, e.j.baurdoux@lse.ac.uk

This dissertation will be mostly based on a recent preprint by Philip Ernst, Alexander L. Stolyar and Jixin Wang (https://arxiv.org/pdf/2404.03797), which concerns an extension of a problem in storage allocation studied in a seminal 1985 paper by Coffman, Kadota, and Shepp (details below).

This leads to a fragmentation problem, which has been studied in a different setting in this 2012 paper https://www.sciencedirect.com/science/article/pii/S0304414911003176

The dissertation would consider whether there is a connection between these two papers, and possible simulation studies to clarify various restuls.

We revisit a classical problem in dynamic storage allocation. Items arrive in a linear storage medium, modeled as a half-axis, at a Poisson rate r and depart after an independent exponentially distributed unit mean service time. The arriving item sizes (lengths) are assumed to be independent and identically distributed (i.i.d.) from a common distribution H. A widely employed algorithm for allocating the items is the "first-fit" discipline, namely, each arriving item is placed in the left-most vacant interval large enough to accommodate it. In a seminal 1985 paper, Coffman, Kadota, and Shepp proved that in the special case of unit length items (i.e. degenerate H), as r tends towards infinity, the first-fit algorithm is asymptotically optimal in the following sense: the steady-state ratio of expected "empty space" (gaps between items) to expected occupied space tends towards 0. In a sequel to [6], Coffman, Kadota, and Shepp ([5]) conjectured that the first-fit discipline is also asymptotically optimal for non-degenerate H.

Prerequisite: ST409

The Geometric Median

Supervisor: Professor Wicher Bergsma, w.p.bergsma@lse.ac.uk

The sample median is a widely used measure of location that is less sensitive to extreme observations than the sample mean. A natural extension to multivariate data is the **geometric median**, also called the spatial median. For observations $x_1, \ldots, x_n \in \mathbb{R}^d$, it is defined by

$$\hat{\mu} = \arg\min_{\mu \in \mathbb{R}^d} \sum_{i=1}^{n} \|x_i - \mu\|_2.$$

Whereas the sample mean minimises the sum of squared Euclidean distances, the geometric median minimises the sum of distances. It reduces to the usual median in one dimension and, unlike taking medians separately for each variable, respects rotations of the coordinate system. Its statistical properties make it an appealing alternative for multivariate data containing outliers (Brown, 1983).

This project will investigate the theoretical properties and practical performance of the geometric median. The main question is how its resistance to outliers compares with its statistical efficiency under different distributions and in different dimensions. Its interpretation as a maximum likelihood estimator under a spherical density proportional to $\exp(-\|x - \mu\|_2/s)$ provides a further connection with familiar methods of statistical inference.

The core of the dissertation will comprise:

- **Theory:** a review of selected properties, including existence, uniqueness, behaviour under transformations and robustness. Established asymptotic results may be used to help explain the simulation findings (Möttönen, Nordhausen and Oja, 2010).
- **Computation:** implementation of an algorithm for calculating the geometric median, such as the modified Weiszfeld algorithm of Vardi and Zhang (2000), with attention to convergence and numerical accuracy.
- **Simulation:** comparison with the sample mean and coordinatewise median under multivariate normal and heavy-tailed distributions, with and without outlier contamination. A focused selection of sample sizes and dimensions will be used to examine estimation error and computational performance.

Depending on the candidate's interests, one extension could investigate the effect of rescaling variables, confidence regions for the population geometric median, or a more detailed comparison of computational algorithms. The balance between theory and computation can be adapted to the candidate. Familiarity with statistical inference, linear algebra and programming in R or Python would be useful.

Suggested references

Brown, B. M. (1983). Statistical uses of the spatial median. *Journal of the Royal Statistical Society, Series B*, 45(1), 25–30.

Vardi, Y. and Zhang, C.-H. (2000). The multivariate L_1-median and associated data depth. *Proceedings of the National Academy of Sciences*, 97(4), 1423–1426.

Möttönen, J., Nordhausen, K. and Oja, H. (2010). Asymptotic theory of the spatial median. *In Nonparametrics and Robustness in Modern Statistical Inference and Time Series Analysis*. IMS Collections, 7, 182–193.

Maximum Product of Spacings Estimation

Supervisor: Professor Wicher Bergsma, w.p.bergsma@lse.ac.uk

Maximum product of spacings estimation is an alternative to maximum likelihood for continuous statistical models. In one dimension, it estimates parameters by maximising the product of the probabilities assigned to intervals between consecutive ordered observations, including the two tail intervals. The method is closely related to likelihood and can remain useful in some models where the likelihood is unbounded. This project will investigate its theoretical foundations and computational performance, with a brief study of an extension to two dimensions.

The main component will focus on the univariate setting. The student will review selected results on consistency and efficiency, derive estimators in simple examples, and implement a comparison with maximum likelihood in R or Python. A focused simulation study will investigate bias, mean squared error and numerical stability in selected models, including one with an unknown endpoint. The uniform distribution with an unknown upper endpoint provides an accessible starting example; a shifted Weibull model offers a more demanding case.

In two dimensions there is no natural ordering of observations. An established extension uses Voronoi cells, each consisting of the points closer to one observation than to any other. Parameters are estimated by maximising the product of the model probabilities assigned to these cells. The student will explain this construction and illustrate it in a small bivariate normal example, estimating one mean component while treating the other mean component and the covariance matrix as known. Standard numerical integration or Monte Carlo methods can be used to approximate cell probabilities, with a check of how numerical accuracy affects the estimates.

The project is intended for a one-unit MSc dissertation. Its contribution will be a clear synthesis of published theory, reproducible implementations and a critical empirical comparison. The univariate analysis will form the core, with the two-dimensional example illustrating the additional geometric and computational difficulties. New multivariate methodology and new asymptotic theory are beyond the intended scope.

References

Cheng, R. C. H. and Amin, N. A. K. (1983). Estimating parameters in continuous univariate distributions with a shifted origin. *Journal of the Royal Statistical Society, Series B*, 45(3), 394–403. doi: 10.1111/j.2517-6161.1983.tb01268.x

Ranneby, B., Jammalamadaka, S. R. and Teterukovskiy, A. (2005). The maximum spacing estimation for multivariate observations. *Journal of Statistical Planning and Inference*, 129(1–2), 427–446. doi: 10.1016/j.jspi.2004.06.059

What's the Harm? Understanding the Distribution of Individual Treatment Effects

Supervisor: Dr. Yining Chen, Y.Chen101@lse.ac.uk

In causal inference the average treatment effect is the standard summary of treatment efficacy, but a positive average can mask the fact that a non-trivial fraction of the population is harmed by treatment. To understand this heterogeneity we need the distribution of the individual treatment effect — the difference between a person's outcome under treatment and under control. The fundamental difficulty is that this quantity is never observed: even in a perfect randomised experiment we can identify the two marginal outcome distributions, but their joint distribution, and hence the distribution of individual effects, depends on the unidentified dependence structure between potential outcomes.

This thesis studies what can be learned about the distribution of individual treatment effects from the two marginals alone, and how additional assumptions sharpen the picture. The starting point is the classical Makarov (1982) bounds, which give the sharp upper and lower envelopes over all joints consistent with the marginals. These bounds, and bounds on derived quantities such as the probability of being harmed and the conditional value at risk, can be formulated as optimal transport problems: minimising or maximising a functional over the set of all couplings of the two marginals, with Kantorovich duality converting the intractable optimisation over unobserved joints into expressions involving only the observable marginals. The thesis will then examine how bounds tighten under stronger assumptions — rank invariance (equivalent to the monotone transport coupling, yielding the quantile treatment effect as point identification), copula models for sensitivity analysis, monotone treatment response, and conditioning on pre-treatment covariates — and develop plug-in estimators with bootstrap inference. A simulation study will assess finite-sample coverage and bound width across scenarios with constant, heterogeneous, and covariate-dependent treatment effects.

References:

Fan, Y. and Park, S. S. (2010). Sharp bounds on the distribution of treatment effects and their statistical inference. Econometric Theory, 26(3), 931–951.

Firpo, S. and Ridder, G. (2019). Partial identification of the treatment effect distribution and its functionals. Journal of Econometrics, 213(1), 210–234.

Kallus, N. (2022). What's the harm? Sharp bounds on the fraction negatively affected by treatment. Advances in Neural Information Processing Systems (NeurIPS), 35. Kallus, N. (2023). Treatment effect risk: bounds and inference. Management Science, 69(8), 4579–4590.

Makarov, G. D. (1982). Estimates for the distribution function of a sum of two random variables when the marginal distributions are fixed. Theory of Probability and its Applications, 26(4), 803–806.

Nonparametric Efficiency Analysis with Non-Euclidean Inputs

Supervisor: Dr. Yining Chen, Y.Chen101@lse.ac.uk

Production frontier analysis compares each decision-making unit (firm, hospital, bank branch) against a best-practice frontier estimated from the sample, and measures inefficiency as distance to that frontier (Farrell, 1957; Sickles and Zelenyuk, 2019). Two traditions dominate: deterministic nonparametric envelopment estimators such as DEA and the Free Disposal Hull, which are consistent but converge slowly (Park et al., 2000), with recent developments including new tests for returns to scale (Kneip et al., 2025); and stochastic frontier models that separate noise from inefficiency through the asymmetry of a composed error. Almost all of this theory assumes inputs are non-negative Euclidean vectors, whereas in modern applications an "input" is increasingly a function (a load profile) or a distribution (loan sizes at a bank branch, claim sizes at an insurer). Current practice reduces such inputs to summary moments, discarding information. Parallel developments in the statistical treatment of distributional data (Petersen and Müller, 2019) suggest a more principled approach is now feasible.

The aim of this dissertation is to review the existing literature of frontier analysis and seek to develop nonparametric frontier estimation when at least one input is genuinely infinite-dimensional.

References:

Farrell, M. J. (1957). The measurement of productive efficiency. Journal of the Royal Statistical Society, Series A, 120(3):253–290.

Kneip, A., Simar, L., and Wilson, P. W. (2025). Conical FDH estimators for testing returns to scale and making inference about changes in productivity. Econometric Reviews, 45(4):482–517.

Park, B. U., Simar, L., and Weiner, C. (2000). The FDH estimator for productivity efficiency scores: Asymptotic properties. Econometric Theory, 16(6):855–877.

Petersen, A. and Müller, H.-G. (2019). Fréchet regression for random objects with Euclidean predictors. Annals of Statistics, 47(2):691–719.

Sickles, R. C. and Zelenyuk, V. (2019). Measurement of Productivity and Efficiency: Theory and Practice. Cambridge University Press.

A Factor Model for Pairwise Comparison Data and Its Statistical Inference

Supervisor: Dr. Yunxiao Chen, y.chen186@lse.ac.uk

This project considers the statistical modeling and inference of pairwise comparison data, which frequently arises in sports analytics, preference learning, crowdsourcing, and educational assessment. Traditional pairwise comparison models, such as the classical Bradley–Terry (BT) model, assume that each item is characterized by a unidimensional skill or preference score. However, real-world comparative preferences are often multi-dimensional, driven by multiple underlying latent features or

factors. In recent work (Pairwise Comparisons without Stochastic Transitivity: Model, Theory and Applications), Lee and Chen (2026) proposed a generalized approximately low-rank framework for pairwise comparisons that relaxes traditional stochastic transitivity assumptions. Building upon this foundation, this project aims to transition from an approximately low-rank representation to an exact low-rank factor model for pairwise comparison data.

In this dissertation, you will:

1. Formulate the exact low-rank factor model for multi-dimensional pairwise comparison data.
2. Design an efficient algorithm for parameter estimation (e.g., via maximum likelihood or spectral methods).
3. Derive theoretical results establishing the convergence rate and asymptotic properties of the estimator.
4. Establish a Central Limit Theorem (CLT) to enable formal statistical inference, such as constructing confidence intervals and testing hypotheses on latent parameters.
5. Apply the proposed model and algorithm to empirical pairwise comparison datasets (e.g., sports match outcomes or user preference choices) to compare its performance against standard Bradley–Terry baselines.

Depending on your technical level, you may complete a subset of these.

Prerequisite:

Familiar with Python or R (coding is required to implement the estimation algorithm and evaluate its empirical performance), and a solid foundation in mathematical statistics, probability theory, and asymptotic theory (e.g., matrix analysis, convergence proofs, and central limit theorems). It also helps if you take ST405.

Reading:

Bradley, R. A., & Terry, M. E. (1952). Rank analysis of incomplete block designs: I. The method of paired comparisons. Biometrika, 39(3/4), 324–345.

Han, R., Ye, R., Tan, C., & Chen, K. (2020). Asymptotic theory of sparse Bradley–Terry model. The Annals of Applied Probability, 30(5), 2491–2515.

Lee, S. M., & Chen, Y. (2026). Pairwise comparisons without stochastic transitivity: Model, theory and applications. Journal of Machine Learning Research, 27(217), 1–49.

Frequentist Estimation of High-dimensional Logistic Factor Models by Pólya-Gamma-based Stochastic Approximation

Supervisor: Dr. Yunxiao Chen, y.chen186@lse.ac.uk

This project investigates the estimation of a logistic factor model, also known as multi-dimensional two-parameter logistic item response theory models. This model is commonly used in psychological and educational measurement, recommendation systems, among others. In this project, we consider the estimation of this model by maximising a marginal likelihood, which assumes that the latent variables (i.e., factors) are random variables and the item-specific parameters as well as other structural parameters of the model as fixed parameters. Maximising the marginal likelihood is computationally a challenge when the latent dimension is high, due to the integrals with respect to the latent variables in the marginal likelihood. Markov chain Monte Carlo (MCMC) stochastic approximation is commonly used for solving this optimisation problem, which combines the stochastic approximation technique (i.e., stochastic gradient descent) with MCMC sampling technique. Existing works used Metropolis-Hastings sampling techniques for the MCMC sampling. However, with the logistic link function in the model, posterior samples of the latent variables may be obtained more efficiently using the Gibbs sampler based on Pólya-Gamma data augmentation. In this project, you will need to develop an algorithm for MCMC stochastic approximation algorithm (i.e., stochastic gradient descent) based on Pólya-Gamma sampling. You can then compare the algorithm with stochastic approximation based on Metropolis-Hastings samplers to study its efficiency, and further apply it to psychological measurement data or recommendation system data (e.g., moving rating).

Prerequisite:

Familiar with Python or R (coding is needed for this project to put the algorithm together), some basic knowledge about Bayesian statistics. It also helps if you take ST405.

Reading:

Polson, N. G., Scott, J. G., & Windle, J. (2013). Bayesian inference for logistic models using Pólya–Gamma latent variables. Journal of the American statistical Association, 108(504), 1339–1349.

Jiang, Z., & Templin, J. (2019). Gibbs samplers for logistic item response models via the Pólya–Gamma distribution: A computationally efficient data-augmentation strategy. Psychometrika, 84(2), 358-374.

Zhang, S., & Chen, Y. (2022). Computation for latent variable model estimation: A unified stochastic proximal framework. psychometrika, 87(4), 1473-1502.

Exact Simulation of Lévy Subordinators: Efficiency, Failure Regimes and Improved Envelopes

Supervisor: Professor Angelos Dassios, (COL.7.14, a.dassios@lse.ac.uk)

Description: Exact (bias-free) simulation of Lévy subordinators and related distributions (tempered stable, truncated stable, generalised gamma, inverse Gaussian, and their bridges) relies heavily on acceptance–rejection schemes built on stable or series representations. These algorithms are exact, but their cost can degrade sharply in certain parameter regimes. Examples include the stability index approaching 1, extreme tempering, and very small or very large time horizons. This project asks the student to understand *why* these methods slow down and to do something about it.

The student will first give a concise, mathematically rigorous account of the main exact-simulation approaches, including Zolotarev/Kanter-type representations, Devroye's schemes for tilted stable laws, and the Dassios–Qu–Zhao and Dassios–Lim–Qu algorithms. They will then implement the methods in R from first principles, without relying on packaged samplers.

The core of the project is an efficiency analysis. The student will derive, analytically where possible, the acceptance probability and expected computational cost of each scheme as a function of the parameters. They will identify asymptotic regimes where performance collapses and confirm these findings with carefully designed numerical experiments.

Finally, the student will propose at least one modification, such as a new envelope, a decomposition of the Lévy measure, or a hybrid scheme switching between methods by regime. They must prove that the modification remains exact and show where it improves on existing methods.

At the start of the project, the student will be assigned a specific subordinator or bridge not treated in full in the standard references. This is negotiable. The final part of the work must be carried out for that case.

As part of assessment, there will be a thorough viva to make sure there was no AI assistance

Expected outputs:

- A derivation of acceptance rates and expected cost for each method, including asymptotics in the critical parameter limits.
- A validated R implementation. Validation must include comparison with the distribution recovered by numerical inversion of the Laplace transform, moment checks, and goodness-of-fit tests.
- Efficiency maps over the parameter space, with a recommended choice of algorithm for each region.
- A proof of exactness and a performance analysis for the proposed improvement.
- A fully reproducible code repository with version history maintained throughout the project.

Prerequisites: Advanced knowledge of R; ST425 and ST460 (both essential).

Preliminary reading:

S. M. Ross, *Simulation*.

L. Devroye (1986), *Non-Uniform Random Variate Generation*, Springer.

L. Devroye (2009), Random variate generation for exponentially and polynomially tilted stable distributions, *ACM Transactions on Modeling and Computer Simulation*.

K.-K. Kim & S. Kim (2016), Simulation of tempered stable Lévy bridges and its applications, *Operations Research*.

A. Dassios, Y. Qu & H. Zhao (2018), Exact simulation for a class of tempered stable and related distributions, *ACM TOMACS*.

A. Dassios, J. W. Lim & Y. Qu (2020), Exact simulation of a truncated Lévy subordinator, *ACM TOMACS*.

Numerical Aspects of Parisian Options: Laplace Inversion, Simulation Bias and Their Reconciliation

Supervisor: Professor Angelos Dassios, (COL.7.14, a.dassios@lse.ac.uk)

Description: Parisian options are knocked in or out only when the underlying spends a continuous period of prescribed length beyond a barrier. Their prices are available in closed form only through Laplace transforms, derived via excursion theory and, more recently, via recursive series for the density of the Parisian stopping time. Both routes are delicate in practice.

Laplace inversion can become unstable when the window is short relative to maturity, when the barrier is close to the spot, or at long maturities. Recursive series can suffer from slow convergence and cancellation. Monte Carlo methods, meanwhile, carry a systematic bias because excursion durations are measured on a discrete grid, and this bias does not vanish at the usual rate.

The project begins with a rigorous synthesis of the theory for single- and double-barrier Parisian options. This covers the excursion-theoretic approach of Chesney, Jeanblanc and Yor, the work of Dassios and Wu, and the recursive formulae of Dassios and Lim. The emphasis is on the structure of the transforms and on where numerical difficulties originate.

The student will implement at least two inversion methods (for example Abate–Whitt's Euler algorithm, Talbot's method and Gaver–Stehfest). They will also implement the recursive density formulae, and will map out accuracy and stability across the parameter space.

In parallel, they will build a Monte Carlo pricer and quantify its discretisation bias both empirically and theoretically. They will then implement and assess a correction, such as Brownian-bridge adjustment of excursion lengths, a continuity correction to the window, or Richardson extrapolation.

The two approaches must be reconciled. The student should produce reference prices with certified error bounds in regimes where the existing literature reports none, and explain every discrepancy that remains.

At the start of the project, the student will be assigned a specific contract variant, for example cumulative (ParAsian), double-barrier with asymmetric windows, or a Parisian feature under a perturbed or regime-switching model. The final analysis must be carried out for that variant, including the derivation of its transform where needed.

As part of assessment, there will be a thorough viva to make sure there was no AI assistance

Expected outputs:

- A self-contained derivation of the Laplace transforms used, for the assigned variant as well as the standard cases.
- Stability and accuracy maps for each inversion method and the recursive formulae, with a diagnosis of each failure mode.
- A theoretical and empirical analysis of Monte Carlo discretisation bias, together with a tested correction.
- A table of benchmark prices with error bounds, cross-validated between methods.
- A fully reproducible code repository with version history maintained throughout the project.

Prerequisites: Very good working knowledge of R; ST425 and ST460 (both essential); ST461 (highly desirable).

Preliminary reading:

S. M. Ross, *Simulation*.

M. Chesney, M. Jeanblanc-Picqué & M. Yor (1997), Brownian excursions and Parisian barrier options, *Advances in Applied Probability*.

A. Dassios & S. Wu (2010), Perturbed Brownian motion and its application to Parisian option pricing, *Finance and Stochastics*.

A. Dassios & S. Wu (2011), Double-barrier Parisian options, *Journal of Applied Probability*.

A. Dassios & J. W. Lim (2013), Parisian option pricing: a recursive solution for the density of the Parisian stopping time, *SIAM Journal on Financial Mathematics*.

J. Abate & W. Whitt (2006), A unified framework for numerically inverting Laplace transforms, *INFORMS Journal on Computing*.

Improving Statistical Inference Using External Predictions

Supervisor: Dr. Oliver Feng, o.feng@lse.ac.uk

In real-world supervised learning problems, time or resource constraints may limit the availability of complete data where both the covariates and responses are recorded, e.g. in image classification where manual labelling by human experts is costly. On the other hand, one may also have access to a black-box machine learning system or external estimator that yields predictions (e.g. artificial image labels) based on covariate input. Given a much larger dataset in which the responses (and possibly also the covariates) are missing, it is natural to ask whether the external predictions can be combined with the fully labelled data to improve statistical estimation and inference. The aim of this project is to explore approaches to this problem that can achieve significant gains, even when the predictions are inaccurate and hence should not be incorporated directly as additional covariates or surrogate responses. Starting with a comprehensive literature review, a student may wish to compare and/or modify existing methodology, conducting theoretical and/or empirical analysis. Another possible direction is to explore how to implement and refine these procedures in a specific application area, e.g. imaging or biostatistics.

References:

[1] A. N. Angelopoulos et al. (2023). Prediction-powered inference. Science 382 (6671), 669–674.

[2] E. Xia and M. J. Wainwright (2024). Prediction aided by surrogate training. arXiv preprint arXiv:2412.09364.

[3] W. J. Underwood et al. (2025). Upgrading survival models with CARE. arXiv preprint 2506.23870.

Score-based Generative Modelling with Diffusion Processes

Supervisor: Dr. Oliver Feng, o.feng@lse.ac.uk

In machine learning, the aim of generative modelling is to obtain artificial samples that are similar (in distribution) to those in a training dataset, which is often a large database of high-dimensional objects such as images or videos. There has recently been a surge of interest in procedures that incrementally add noise to the training examples and continually learn the *score function* of the noisy data, namely the derivative of its log-likelihood. The estimated scores are then used to transform pure noise back into a new sample, via Langevin dynamics or more generally a diffusion process defined by a stochastic differential equation. Neural network implementations of scorebased algorithms have achieved remarkable empirical success, often outperforming other state-of the art methods like generative adversarial networks.

The goal of the project is to explore this rapidly growing field, which has gained significant traction not only in the artificial intelligence community but also among statisticians seeking theoretical guarantees. The student is expected to give a detailed mathematical exposition of score-based generative modelling, including the fundamentals of *score matching* as a key ingredient. One

possible follow-up direction is to apply and/or adapt the methodology to a generative modelling task of the student's choice, using simulated or real data. It may be of interest to investigate extensions of score matching to data on different domains. An alternative option is to survey and perhaps to extend some recent mathematical results on diffusion processes and their convergence properties, or the accuracy of the samples generated in a nonparametric context.

References:

1. Y. Song (2021). Generative modelling by estimating gradients of the data distribution (and references therein). https://yang-song.net/blog/2021/score/
2. G. Li et al. (2023). Towards faster non-asymptotic convergence for diffusion-based generative models. arXiv preprint arXiv:2306.09251.
3. Z. Dou et al. (2024). From optimal score matching to optimal sampling (and references therein). arXiv preprint arXiv:2409.07032.
4. https://scorebasedgenerativemodeling.github.io/

Foundation Models for Tabular Data – Do They Deserve the Hype?

Supervisor: Professor Piotr Fryzlewicz, P.Fryzlewicz@lse.ac.uk

Tabular data is, and has always been, the cornerstone of data science. A typical task is to predict the values of one of the columns – the response – using the values of some of the remaining columns – the predictors. In "traditional" machine learning, this involved learning the appropriate predictive function from the training data, on which the values of the response are known, and then using the estimated predictive function on new test data, for which the response is not available and must be predicted. Tabular foundation models promise to do for tabular data what large language models do for text, which is to use one of the 'pre-learned' predictive functions, trained on a large corpus of synthetic prediction problems and selected at inference time based on the samples from the current dataset.

The objectives of the project will be as follows:

- To understand the exact mechanics of TabPFN, a market-leading tabular foundation model,
- To perform numerical experiments to understand the practical performance of TabPFN in various contexts,
- To understand how the approach differs from typical statistical and machine learning approaches, and its advantages and limitations with respect to those,
- To apply TabPFN or its variants to a challenging modern dataset of interest, e.g. involving weather data or similar.
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(Note: this will be an "AI-responsible" project in the sense that the student will only be permitted to include in the dissertation material that has been thoroughly discussed in meetings, to ensure that no written text that is not fully understood by the student appears in the dissertation.)

Preliminary reading: https://www.nature.com/articles/s41586-024-08328-6

Please be advised: Due to other commitments this term, Professor Fryzlewicz is available to supervise one student on this topic, with regular supervision starting in January 2027. Interested students should first contact Dr. Yining Chen (Y.Chen101@lse.ac.uk), who will explain the arrangement and support the early stages, including the one-page proposal.

Boundary Attainment of Multivariate Markov Processes

Supervisor: Dr. David Itkin, D.Itkin@lse.ac.uk

The question of a whether a stochastic processes attains the boundary of its domain is important in many disciplines; e.g. bankruptsy of firms, overcapacity of a queuing system, etc. For Markov processes (outside of the one dimensional case) techniques to establish that a process *does not* hit the boundary are more developed than those to establish that a process *does* hit the boundary of its domain. Recently, test function methods utilizing the (candidate) invariant measure of the process have been used to show boundary attainment occurs in certain cases. The goal of this project would be to study the more general applicability of these methods and its consequences.

Prerequisite: ST409

Indicative reading:

Khasminskii, Rafail. *Stochastic stability of differential equations*. Springer, 2012.

Itkin, David, and Martin Larsson. "Open markets and hybrid Jacobi processes." *The Annals of Applied Probability* 34.3 (2024): 2940-2985

Simulation Schemes for Multivariate Reflected Stochastic Differential Equations

Supervisor: Dr. David Itkin, D.Itkin@lse.ac.uk

Reflected stochastic differential equations model random processes. By construction, these processes stay in the domain of interest, because when the processes reaches the boundary it "bounces" off the boundary and reflects inwardly. This feature makes multivariate reflected processes difficult to simulate efficiently. Recent advances in representing reflected processes as functions of standard processes open the door to more efficient simulation schemes, which exploit these representations. This project with explore the development of such schemes as well as their numerical implementation.

Suggested reading:

Pilipenko, Andrey. An introduction to stochastic differential equations with reflection. Universitätsverlag Potsdam, 2014.

Itkin, David, and Karatzas, Ioannis. "Folding representations of reflected diffusions." arXiv preprint arXiv:2607.28004 (2026).

Gradient Free Inference for Bayesian PDE-based Inverse Problems

Supervisor: Dr. Ieva Kazlauskaite, i.kazlauskaite@lse.ac.uk

In this project, we will consider Bayesian inference in partial differential equation (PDE)-based inverse problems where we try to estimate PDE parameters from observations of the PDE solution. Specifically, we will focus on Bayesian inference methods that do not use gradient information as acquiring gradients for a PDE solution with respect to the parameters of interest is typically a challenging scientific computing task.

This project will involve:

- Conducting a literature review on gradient-free Bayesian methods, including simulationbased inference and Bayesian optimisation.
- Learning about PDE-based inverse problems, focusing on simple problems such as the steady-state heat equation.
- Implementing simple inference algorithms, such as the Metropolis–Hastings algorithm, simulation-based inference, and Bayesian optimization.

An implementation of a PDE solver for a steady state heat equation will be provided.

Prerequisites:

Proficiency in Python is essential. ST308 and ST303 are recommended, while MA209 and/or MA319 are optional

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Improved Schemes for Parameter Estimation in Statistical Finite Elements Models

Supervisor: Dr. Ieva Kazlauskaite, i.kazlauskaite@lse.ac.uk

In this project, we will consider a new scheme for estimation of parameters of partial differential equations (PDEs) following the framework proposed in [1]. We will explore the advantages and drawbacks of using a Metropolis adjusted Langevin algorithm (MALA) as opposed to the unadjusted LA which introduces discretisation bias. Specifically, we will examine whether the

Metropolis version of the scheme used in [1] leads to a more stable algorithm for fitting the PDE parameters.

This project will involve:

- Conducting a literature review on PDE-based inverse problems and their solution within the statistical finite element model [2].
- Understanding the algorithms and implementation of the ULA algorithm for the statistical finite element model as given in [1].
- Implementing a Metropolis adjusted Langevin algorithm for inference in the framework presented in [1] and conducting an empirical study comparing the results for the two algorithms.

Prerequisites:

Proficiency in Python is essential. ST308 and ST303 are recommended, while MA209 and/or MA319 are optional.

Reading:

[1] Glyn-Davies, A., Duffin, C., Kazlauskaite, I., Girolami, M., & Akyildiz, Ö. D. (2024). Statistical Finite Elements via Interacting Particle Langevin Dynamics. arXiv preprint arXiv:2409.07101.

[2] Girolami, M., Febrianto, E., Yin, G., & Cirak, F. (2021). The statistical finite element method (statFEM) for coherent synthesis of observation data and model predictions. Computer Methods in Applied Mechanics and Engineering, 375, 113533.

Binary Logistic Regression with a Misclassified Response

Supervisor: Professor Jouni Kuha (CON.2.05, j.kuha@lse.ac.uk)

Suppose that a logistic regression model holds for a binary response variable Y but instead of Y we observe a Y* which is a misclassified version of Y (i.e. for some units Y=0 but Y*=1, or vice versa). If Y* is simply used in place of Y, estimated coefficients of the model will be biased. This bias can be corrected by using adjusted methods of estimation. The choice of such methods depends on, among other things, what kind of information is available on the misclassification model between Y and Y*.

This project is likely to involve

- examination of the biases which arise from ignoring the misclassification, using both general theoretical results and simulation studies
- review and comparison of different estimators to correct for the bias
- evaluation of the behaviour of different estimators in different circumstances, using simulation studies
- illustration of the methods using real data

Prerequisites: ST411 and ST405 will be useful.

Indicative readings:

Neuhaus, J. M. (1999). Bias and efficiency loss due to misclassified responses in binary regression. *Biometrika*, 86, 843-855.

Bakk, Z., Tekle, F. T., & Vermunt, J. K. (2013). Estimating the association between latent class membership and external variables using bias-adjusted three-step approaches. *Sociological Methodology*, 43, 72-311.

Bakk, Z. and Kuha, J. (2018). Two-step estimation of models between latent classes and external variables. *Psychometrika*, 83, 871-892.

Two-step Estimation of Structural Equation Models

Supervisor: Professor Jouni Kuha (CON.2.05, j.kuha@lse.ac.uk)

Bakk and Kuha (2018) and Rosseel and Loh (2023) have described different instances of the method of `two-step' estimation for latent variables models. They argue that it shares the good properties of alternative `one-step' and `three-step' methods, and avoids their conceptual and practical disadvantages. In this project you will examine similar methods for structural equation models where the observed and latent variables are continuous. Depending on your preferences for the focus of the dissertation, the work will involve theoretical examination of the properties of the estimators, simulation studies and/or examples with real data.

Data sources: Data for illustrative examples will be easy to obtain.

Prerequisites: ST405

Indicative readings:

Bakk, Z. and Kuha, J. (2018). Two-step estimation of models between latent classes and external variables. *Psychometrika*, 83, 871-892.

Rosseel, Y. and W. W. Loh (2024). A structural after measurement approach to structural equation modeling. *Psychological Methods* 29, 561–588.

Mean Field Games and Mean Field Control for Green Finance

Supervisor: Dr. Giulia Livieri, g.livieri@lse.ac.uk

Mean Field Games and Mean Field Controls can be used to analyze various situations in the context of Green Finance. An example can be the analysis of the electricity producers' decision on how much renewable sources of production ought to be used in the presence of a carbon tax, when the number of electricity producers becomes very large. Different policy scenarios can be considered.

This project will involve the following steps:

a) Review of some stochastic models usually described by a Stochastic Differential Equation (SDE).
b) Review of Mean Field Games and Mean Field Control in general.
c) Review of Mean Field Games and Mean Field Control with a special focus to green finance.
d) A possible simulation part with various policy scenarios.

Prerequisite: ST409

Tentative Bibliography:

R. Carmona, F. Delarue. (2018). Probabilistic theory of mean field games with applications I-II. Springer.

R. Carmona. (2020). Applications of Mean Field Games in financial engineering and economic theory. arXiv:2012.05237.

R. Carmona, G. Dayanikli, M. Lauriere. (2022). Mean Field Models to Regulate Carbon Emissions in Electricity Production. Dynamic Games and Applications. 12:897-928.

Variables Selection via Variational Bayes

Supervisor: Dr. Giulia Livieri, g.livieri@lse.ac.uk

Variational Bayesian methods are a family of techniques for approximating intractable integrals arising in Bayesian inference and machine learning. In particular, Variational Bayes methods are primarily used to derive the so-called lower bound for the marginal likelihood of the observed data. You will use the latter for performing model selection. Different model selection techniques will be explored, both theoretically and numerically.

This project will involve the following steps:

- Review of Variational Bayes methods for variables selections.
- Possible extension of existing methods.

- Numerical implementation (both on simulated and real data).

Prerequisite: ST425 or Equivalent(s)

Tentative Bibliography:

J.T. Ormerod, C. You, S. Muller. (2017). A variational Bayes approach to variable selection.

Electronic Journal od Statistics 11(2):3549-3594

Spike-and-slab meets LASSO: A review of the spike-and-slab LASSO. (2021). Handbook of Bayesian variable selection. 81-108.

Reinforcement Learning Based Parametric Portfolio Policies

Supervisor: Dr. Giulia Livieri, g.livieri@lse.ac.uk

In a very recent strands of literature, researchers have consider parametric portfolio policies of any complexity using deep neural networks to optimize investor utility. The idea of the thesis is to investigate if the problem can be solved via a reinforcement learning approach, both theoretically and empirically.

Main reading:

Simon, Frederik, Sebastian Weibels, and Tom Zimmermann. "Deep parametric portfolio policies." *Management Science* (2026).

Variance-reduced Alignment for Large Language Models

Supervisor: Dr. Chengchun Shi c.shi7@lse.ac.uk

Reinforcement learning from human feedback (RLHF) has become a gold standard for aligning large language models (LLMs) with human preferences. However, most existing RLHF algorithms rely on the Bradley–Terry (BT) model, which makes unrealistic assumptions about human preferences that often fail to capture their complexity and variability in real-world settings.

Ye et al. (2025) recently proposed variance-reduced preference optimization (VRPO; see references), an algorithm designed to improve the sample efficiency and robustness of existing RLHF methods under model misspecification. They demonstrated VRPO's strong empirical performance on the IMDb, Helpful–Harmless (HH), and TL;DR datasets.

This project aims to further advance VRPO by improving its computational and statistical efficiency and by extending its application to the UltraFeedback dataset, which is more complex than the three aforementioned datasets in both data scale and response complexity. The project will be codingintensive. Its successful completion is expected to lead to a research paper suitable for submission to a (top-tier) machine learning conference or statistical journal.

References:

https://arxiv.org/abs/2504.03784

https://arxiv.org/pdf/2305.18290

https://arxiv.org/pdf/2203.02155

Semi-supervised Reinforcement Learning from Human Feedback for LLM Fine-tuning

Supervisor: Dr. Chengchun Shi c.shi7@lse.ac.uk

We have entered an era of large language models (LLMs), which have rapidly transformed how we learn, work and communicate with each other. These models, with billions of parameters, can generate remarkably human-like text. Yet, despite their impressive capabilities, ensuring that LLMs behave in ways that align with human values – being helpful, honest, and safe – remains one of the most urgent challenges in generative artificial intelligence.

To address this, researchers commonly apply reinforcement learning from human feedback (RLHF) to fine-tuning LLMs based on human preferences among multiple responses. However, this approach depends heavily on expensive human feedback, often resulting in small labeled datasets. This data scarcity can lead to inefficient training and unpredictable model behavior.

This project studies semi-supervised RLHF — an approach that integrates semi-supervised learning (SSL) to enhance existing RLHF algorithms by using both labeled and unlabeled data. While LLMs are typically trained with massive amounts of unlabeled text, to our knowledge, SSL has rarely been applied to RLHF. Building on recent advances in SSL, this project will leverage the rich information in unlabeled data to improve LLM's alignment with human values. The project will be coding-intensive. Its successful completion is expected to lead to a research paper suitable for submission to a (toptier) machine learning conference or statistical journal.

References:

https://arxiv.org/pdf/2502.17741

https://www.science.org/doi/10.1126/science.adi6000

https://arxiv.org/abs/2504.03784

https://arxiv.org/pdf/2305.18290

https://arxiv.org/pdf/2203.02155

Unit Roots and Functional Convergence of Stochastic Integrals

Supervisor: Dr. Andreas Søjmark, a.sojmark@lse.ac.uk

This is a theoretical project at the interface of statistical theory and stochastic analysis. It is essential to have a good background in stochastic integration and ideally some understanding of functional analysis to get up to speed on relevant functional topologies and corresponding notions of convergence. As a starting point, the project will explore the asymptotic theory behind unit root tests in moving averages (relevant, e.g., for testing cointegration), as studied in Davis and Song (2011). Since these tests are based on maximum likelihood, the theory already has a functional aspect: the limits of the estimators follow from weak convergence of the likelihood, and thus of the underlying stochastic integrals, as functions of the parameters. The existing theory relies on finite variance assumptions, which lead to limits driven by Brownian motion. This project would aim to develop results for heavy-tailed settings where the innovations can have infinite variance. In such settings, the functional convergence of partial sums of moving averages becomes much more intricate. Moreover, the project could explore other functional aspects, where pathwise properties of the time series become important for the stability of estimators, for example due to change points or time-varying regression coefficients. This calls for functional convergence jointly in time and in the parameters, which would involve more delicate use of the machinery for functional convergence of stochastic integrals.

Prerequisites: ST409, ST460

Indicative reading:

- Davis and Song (2011): Unit roots in moving averages beyond first order, Annals of Statistics
- Davis and Song (2012): Functional convergence of stochastic integrals with application to statistical inference, Stochastic Processes and their Applications
- Sojmark and Wunderlich (2026): Weak convergence of stochastic integrals on Skorokhod space in Skorokhod's J1 and M1 topologies, Probability Theory and Related Fields 195, 997–1049

Epidemic models with diffusive heterogeneity

Supervisor: Dr. Andreas Søjmark, a.sojmark@lse.ac.uk

This is a project at the interface of probabilistic modelling, differential equations and computational statistics. It concerns epidemic models in which heterogeneity in the population is captured through diffusive dynamics, in contrast to the fixed, homogeneous contact and transmission rates of the classical SIR model. The project will begin by laying out the mathematical theory. On one side, the models are formulated at the level of individuals as interacting stochastic differential equations. On the other side, one can study reaction–diffusion equations and the partial differential equations arising as large-population limits of interacting systems. The project would then aim to calibrate these models to

data from past epidemics, assessing their performance and comparing the models. Moreover, one could explore the modelling of interventions or examine the identifiability of the model parameters from aggregate infection data.

Prerequisites: Ideally ST409, or at least some background in stochastic processes. Ideally also some background in ordinary and partial differential equations. Coding in Python.

Indicative reading:

- Berestycki, H., Desjardins, B., Weitz, J. S. and Oury, J.-M. (2023). Epidemic modeling with heterogeneity and social diffusion. Journal of Mathematical Biology 86, 60.
- Fausti, E. and Søjmark, A. (2022). An interacting particle system for the front of an epidemic advancing through a susceptible population. arXiv:2210.09286.
- Fausti, E. and Søjmark, A. (2026). A free boundary problem for the mean-field limit of diffusing particles with nonlinear boundary reactivity. arXiv:2604.12797.

Selection or Influence? A Comparison of Longitudinal Models for Dyadic Panel Data, with an Application to Couples' Mental Health

Supervisor: Professor Fiona Steele, f.a.steele@lse.ac.uk

Previous research suggests that there is a strong interdependence in couples' health behaviours and outcomes. Possible explanations are selection of individuals with similar characteristics (including attitudes and behaviours) into couples, and mutual effects of each partner's health on the other's.

The project involves the design and conduct of simulation studies and the analysis of panel data on mental health to answer three linked questions:

- How well do commonly used longitudinal models separate stable similarity between partners (selection) from lagged partner influence?
- Under what realistic conditions do they produce spurious or biased partner effects?
- Do the substantive conclusions about couples' mental health in the UK change depending on which model is used?

Data:

UK Household Longitudinal Survey (aka Understanding Society). For details see https://www.understandingsociety.ac.uk/ Note that this project will require a large amount of time devoted to preparing the data for analysis.

Prerequisites:

ST442 an advantage

Reading:

Kenny, D. A., Kashy, D. A., & Cook, W. L. (2006). *Dyadic data analysis*. New York: Guilford Press.

Hamaker, E. L., Kuiper, R. M., & Grasman, R. P. P. P. (2015). A critique of the cross-lagged panel model. *Psychological Methods*, *20*(1), 102–116. https://doi.org/10.1037/a0038889

Analysis of Clustered Social Network Data

Supervisor: Professor Fiona Steele, f.a.steele@lse.ac.uk

The social relations model (SRM) is widely used to analyse social network data where interest lies in the predictors of the probability of a tie between a pair of individuals (referred to as a dyad). Ties may be directed (e.g. provision and receipt of support from each individual to the other) or undirected. The SRM can be framed as a non-hierarchical random effects model where each individual is a member of multiple dyads and was extended to handle clustered data where ties are restricted to individuals in the same cluster (Snijders & Kenny 1999). This project considers models for the situation where individuals are nested in household clusters and there are ties between individuals from different households.

Data:

Data on exchanges of various kinds of support between individuals in 75 villages in Karnataka, India. The data can be downloaded from https://web.stanford.edu/~jacksonm/Data.html (a zip file containing all raw data is under 'Data on Social Networks ...'). Other suitable social network datasets could be analysed.

Prerequisites:

- ST411 would be helpful, but not essential
- Good programming skills to prepare data for analysis
- A willingness to work out how to estimate models using MCMC methods via the rjags or rstan R packages

Reading:

Hoff, P. D. (2005). Mixed-effects models for dyadic data. *Journal of the American Statistical Association*, *100*(469), 286–295. https://doi.org/10.1198/016214504000001015

Steele, F., Weltz, J., Koster, J. and E. Power (2026) Multilevel modelling of double-sampled clustered social networks with individual-level data on between-cluster ties. arXiv:2608.26387

Minimax Estimation of Kernel-based Information Theoretical Measures

Supervisor: Professor Zoltan Szabo, z.szabo@lse.ac.uk

Topic: Kernel-based techniques [1] are at the forefront of estimating information theoretical quantities including divergence, independence and goodness-of-fit measures such as maximum mean discrepancy (MMD; [2]), Hilbert-Schmidt independence criterion (HSIC, [3]) and kernel Stein discrepancy (KSD [4,5]), thanks to the flexibility of the underlying reproducing kernel Hilbert space (RKHS) under mild conditions expressed by the characteristic [6,7] or universal property [8,9] of the kernel. Despite the wide range of applications of these information theoretical measures, the fundamental question of the fastest rate at which these quantities can be estimated (known as minimax rate [10]) has only quite moderately been understood [11-13]. The goal of this project is to (i) explore the minimax estimation of kernel-based information theoretical measures, (ii) relax existing strong assumptions imposed (such as translation invariance on R^d), (iii) extend the results to new measures such as kernel Lancaster and Streitberg interaction [14], and kernel cumulants [15].

Prerequisites: Solid statistical knowledge is required; familiarity with information theory, functional analysis and measure theory are beneficial.

References:

[1] Ingo Steinwart and Andreas Christmann. Support Vector Machines. Springer, 2008.
[2] Arthur Gretton, Karsten Borgwardt, Malte Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. Journal of Machine Learning Research, 13(25):723–773, 2012.
[3] Arthur Gretton, Olivier Bousquet, Alex Smola, and Bernhard Schölkopf. Measuring statistical dependence with Hilbert-Schmidt norms. In Algorithmic Learning Theory (ALT), pages 63–78, 2005.
[4] Qiang Liu, Jason Lee, and Michael Jordan. A kernelized Stein discrepancy for goodness-of-fit tests. In International Conference on Machine Learning (ICML), pages 276–284, 2016.
[5] Kacper Chwialkowski, Heiko Strathmann, and Arthur Gretton. A kernel test of goodness of fit. In International Conference on Machine Learning (ICML), pages 2606–2615, 2016.
[6] Kenji Fukumizu, Arthur Gretton, Xiaohai Sun, and Bernhard Schölkopf. Kernel measures of conditional dependence. In Advances in Neural Information Processing Systems (NIPS), pages 498–496, 2008.
[7] Bharath Sriperumbudur, Arthur Gretton, Kenji Fukumizu, Bernhard Schölkopf, and Gert Lanckriet. Hilbert space embeddings and metrics on probability measures. Journal of Machine Learning Research, 11:1517–1561, 2010.
[8] Charles Micchelli, Yuesheng Xu, and Haizhang Zhang. Universal kernels. Journal of Machine Learning Research, 7:2651–2667, 2006.
[9] Bharath Sriperumbudur, Kenji Fukumizu, and Gert Lanckriet. Universality, characteristic kernels and RKHS embedding of measures. Journal of Machine Learning Research, 12:2389–2410, 2011.
[10] Alexandre B. Tsybakov. Introduction to Nonparametric Estimation. Springer, 2009.
[11] Ilya Tolstikhin, Bharath Sriperumbudur, and Bernhard Schölkopf. Minimax estimation of maximal mean discrepancy with radial kernels. In Advances in Neural Information Processing Systems (NIPS), pages 1930–1938, 2016.

[12] Florian Kalinke and Zoltán Szabó. The minimax rate of HSIC estimation for translation-invariant kernels. In Advances in Neural Information Processing Systems (NeurIPS), pages 108468–108489, 2024.
[13] Jose Cribeiro-Ramallo, Agnideep Aich, Florian Kalinke, Ashit Baran Aich, and Zoltán Szabó. The minimax lower bound of kernel Stein discrepancy estimation. In International Conference on Artificial Intelligence and Statistics (AISTATS), pages 901-909, 2026.
[14] Dino Sejdinovic, Arthur Gretton, and Wicher Bergsma. A kernel test for three-variable interactions. In Advances in Neural Information Processing Systems (NIPS), pages 1124–1132, 2013.
[15] Patric Bonnier, Harald Oberhauser, and Zoltán Szabó. Kernelized cumulants: Beyond kernel mean embeddings. In Advances in Neural Information Processing Systems (NeurIPS), pages 11049–11074, 2023.

Spectral Distribution Regression

Supervisor: Professor Zoltan Szabo, z.szabo@lse.ac.uk

Description: Distribution regression (DR, [1]) refers to the recent learning paradigm where the inputs in regression are probability distributions or in practice bags of points. Such a bag-of-feature representation is popular in a large number of applications including image processing (bag of visual features), natural language processing (bag of words), aerosol prediction (multi-spectral satellite measurements over an area), or astronomy (galaxy clusters). By using rich feature representations associated to reproducing kernels called mean embedding [2] one can tackle the DR problem by simple ridge regression techniques with minimax-optimal guarantees [3-4] and avoiding density estimation. Unfortunately, the computational complexity of these methods can be prohibitive, cubic in the number of bags with very few techniques possessing reduced workload [5]. This bottleneck motivates the exploration of spectral algorithms in DR which often benefit from adaptivity and reduced memory footprint [6-8]. The goal of this project is to investigate the computational and statistical properties of spectral methods in DR.

Prerequisites: Statistics and programming knowledge (for instance, in Python) are required; background on functional analysis is beneficial.

References:

[1] Barnabás Póczos, Alessandro Rinaldo, Aarti Singh, and Larry Wasserman. Distribution-free distribution regression. International Conference on Artificial Intelligence and Statistics (AISTATS), pages 507-515, 2013.
[2] Krikamol Muandet, Kenji Fukumizu, Bharath Sriperumbudur, and Bernhard Scholkopf. Kernel mean embedding of distributions: A review and beyond. Foundations and Trends in Machine Learning, 10(1-2):1-141, 2017.
[3] Zoltán Szabó, Bharath K. Sriperumbudur, Barnabás Póczos, Arthur Gretton. Learning Theory for Distribution Regression. Journal of Machine Learning Research, 17(152):1-40, 2016.
[4] Zhiying Fang, Zheng-Chu Guo, and Ding-Xuan Zhou, Optimal learning rates for distribution regression, Journal of Complexity 56 (2020), 101426.

[5] Nicole Mücke. Stochastic Gradient Descent Meets Distribution Regression. In International Conference on Artificial Intelligence and Statistics (AISTATS), pages 2143-2151, 2021.
[6] Andreas Neubauer, Heinz Werner Engl, and Martin Hanke. Regularization of Inverse Problems. Springer, 1996.
[7] Gilles Blanchard and Nicole Mücke. Optimal rates for regularization of statistical inverse learning problems. Foundations of Computational Mathematics, 18(4):971-1013, 2018.
[8] Junhong Lin, Alessandro Rudi, Lorenzo Rosasco, and Volkan Cevher. Optimal rates for spectral algorithms with least-squares regression over Hilbert spaces. Applied and Computational Harmonic Analysis, 48(3):868-890, 2020.

How Multicalibrated Is a Model? Valid Confidence Intervals for the Multicalibration Error

Supervisor: Professor Milan Vojnović, m.vojnovic@lse.ac.uk

Abstract:

A predictive model is multicalibrated if its probability estimates are accurate (calibrated) within every group of a specified collection of possibly overlapping subpopulations, such as patients grouped by age, sex or condition; among cases given a probability of 70%, for instance, about 70% should occur. A common way of summarising this is the largest calibration gap over the groups and over bands of predicted probability, restricted to groups above a stated minimum size; this is the target of the project. The collection of groups is taken as given and may be very large, with an extension to groups defined by a rich function class. Because the largest of many estimated gaps is chosen using the same data that estimate it, it is biased upwards, and a confidence interval (a range meant to contain the true value with a stated probability, called its coverage) that ignores this selection may fall short of its stated coverage. Two existing approaches address related problems. Bootstrap-based bounds that hold for every group at once, with asymptotically exact coverage, have been developed for auditing model performance over groups, with multicalibration as an application. Separately, inference for the largest or smallest value of a collection of estimated quantities has been developed in econometrics, with a selection step that shortens the interval. Both guarantees are asymptotic. This project asks, for the worst-group calibration gap, how far an interval can be shortened below the one obtained from bounds on every group at once, by using the fact that the target is a maximum and the structure of the groups, such as nesting and overlap, while keeping its stated coverage. The outcome is a single result: either such an interval, with a proof of its coverage and a comparison of its width with the all-groups interval and with sample splitting (choosing the worst group on one part of the data and estimating its error on the other), or a proof that, for a stated group structure, no valid interval is substantially shorter. Deliverables may also include an open-source release of the proposed method. The work combines proof and simulation, and suits a student with a strong grounding in probability and asymptotic statistics and an interest in the statistical evaluation of machine-learning models.

Indicative readings:

Hébert-Johnson, U., Kim, M. P., Reingold, O. and Rothblum, G. Multicalibration: calibration for the (computationally-identifiable) masses. *Proceedings of the 35th International Conference on Machine Learning (ICML)*, PMLR 80: 1939–1948, 2018.

Guy, I., Haimovich, D., Linder, F., Okati, N., Perini, L., Tax, N. and Tygert, M. Measuring multi-calibration. arXiv:2506.11251.

Cherian, J. J. and Candès, E. J. Statistical inference for fairness auditing. *Journal of Machine Learning Research* 25: 1–49, 2024.

Chernozhukov, V., Lee, S. and Rosen, A. M. Intersection bounds: estimation and inference. *Econometrica* 81(2), 2013.

What Does a Human-Preference Leaderboard Actually Rank?

Supervisor: Professor Milan Vojnović, m.vojnovic@lse.ac.uk

Abstract:

Leaderboards built from human votes between pairs of model responses, such as Chatbot Arena (now LMArena), are widely used to compare models and to inform decisions about their development and adoption. Rankings are computed by pooling votes from a self-selected crowd of users into a single score for each model. In effect, a model's score reflects how often it wins against the other models it is shown against, averaged over voters. If voters disagree, however, the resulting ranking need not match what any particular group of users would conclude, and it may depend on which pairs of models happen to be shown. This project asks what such a ranking actually measures and which ranking claims can be trusted for other user populations. The student will characterise what the pooled ranking estimates when voters are heterogeneous, derive bounds on how a model's rank could change under different or uncertain mixes of users, and develop confidence statements of the form 'model A ranks above model B for every plausible user mix'. The methods will be checked by simulation, applied to publicly released leaderboard votes using language and prompt type as observable user segments, and tested on a dataset in which the voter groups are known. The work combines theory, simulation and data analysis, and suits a student with a strong grounding in likelihood theory and an interest in the statistical evaluation of AI systems.

Indicative readings:

Chiang, W.-L., Zheng, L., Sheng, Y., Angelopoulos, A. N., Li, T., Li, D., Zhu, B., Zhang, H., Jordan, M. I., Gonzalez, J. E. and Stoica, I. (2024). Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference. *Proceedings of the 41st International Conference on Machine Learning (ICML)*, PMLR 235: 8359–8388.

Ameli, S., Zhuang, S., Stoica, I. and Mahoney, M. W. (2025). A Statistical Framework for Ranking LLM-based Chatbots. *International Conference on Learning Representations (ICLR)*.

Siththaranjan, A., Laidlaw, C. and Hadfield-Menell, D. (2024). Distributional Preference Learning: Understanding and Accounting for Hidden Context in RLHF. *International Conference on Learning Representations (ICLR)*.

Kirk, H. R., Whitefield, A., Röttger, P., Bean, A., Margatina, K., Mosquera-Gomez, R., Ciro, J., Bartolo, M., Williams, A., He, H., Vidgen, B. and Hale, S. A. (2024). The PRISM Alignment Dataset: What Participatory, Representative and Individualised Human Feedback Reveals About the Subjective and Multicultural Alignment of Large Language Models. *Advances in Neural Information Processing Systems 37 (NeurIPS), Datasets and Benchmarks Track*.

Frauen, D., Deviyani, A., van der Schaar, M. and Feuerriegel, S. (2026). Nonparametric LLM Evaluation from Preference Data. *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, PMLR 306: 31575–31617.

Conformal Prediction under Distribution Shift

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Conformal prediction provides distribution-free prediction sets with finite-sample marginal coverage guarantees under exchangeability (Vovk et al., 2005). However, this assumption is often violated in practice: the population encountered after deployment may differ from that used to train and calibrate the model. Standard conformal intervals can consequently lose their coverage guarantees.

Recent methods address this problem under different assumptions about the shift. Under covariate shift, importance-weighted conformal prediction reweights calibration observations to represent the target population (Tibshirani et al., 2019). Other approaches provide guarantees under non-exchangeable data, use doubly robust calibration, or protect against distributions in a specified neighbourhood of the training distribution (Barber et al., 2023; Yang et al., 2024; Cauchois et al., 2024). Important practical questions concern estimating importance weights, limited overlap between populations, and the trade-off between coverage and interval width.

In the project, the student will review conformal prediction under distribution shift, implement and compare standard, importance-weighted and robust methods through simulations and real datasets. Simulations may vary the severity of covariate shift, sample size, heteroscedasticity and source–target overlap. Capable students may explore weight truncation, doubly robust calibration, or departures from the covariate-shift assumption.

Prerequisites:

- A solid background in mathematical statistics, including regression, hypothesis testing and asymptotic ideas.

- Good computational skills in R or Python, including implementing algorithms, running simulations and analysing real datasets.
- Optional but desirable: prior exposure to statistical learning, importance sampling or semiparametric methods.

References:

Barber, R. F., Candès, E. J., Ramdas, A., & Tibshirani, R. J. (2023). Conformal prediction beyond exchangeability. The Annals of Statistics, 51(2), 816–845.

Cauchois, M., Gupta, S., Ali, A., & Duchi, J. C. (2024). Robust validation: Confident predictions even when distributions shift. Journal of the American Statistical Association, 119(548), 3033–3044.

Tibshirani, R. J., Barber, R. F., Candès, E. J., & Ramdas, A. (2019). Conformal prediction under covariate shift. Advances in Neural Information Processing Systems, 32.

Vovk, V., Gammerman, A., & Shafer, G. (2005). Algorithmic learning in a random world. Springer.

Yang, Y., Kuchibhotla, A. K., & Tchetgen Tchetgen, E. J. (2024). Doubly robust calibration of prediction sets under covariate shift. Journal of the Royal Statistical Society: Series B, 86(4), 943–965.

Estimating a Target Coefficient with High-dimensional Nuisance

Supervisor: Professor Tengyao Wang, t.wang59@lse.ac.uk

In many modern applications, researchers are interested in a single regression coefficient while controlling for a large number of covariates. Consider the partially linear model

$$y_i = \alpha z_i + \beta^\top x_i + \epsilon_i, \quad i = 1, \dots, n,$$

where we are interested in estimating a single univariate parameter α (e.g. treatment effect) in the presence of high-dimensional nuisance parameter β (e.g. other demographic covariates). Standard Lasso estimates are biased, making inference on α challenging. In this project, we look at several approaches developed to address this problem: including the 'partialling-out' method of Robinson (1988), the debiased Lasso approach of Zhang and Zhang (2014) and double machine learning method of Chernozhukov et al. (2018). The student will review the relevant theory, implement these estimators in R/Python, and conduct simulations comparing their finite-sample coverage, bias, and power across different regimes ($p < n$, $p \gg n$, correlated designs, heteroscedasticity).

Prerequisite:

- A solid background in mathematical statistics (including high-dimensional linear model, likelihood-based inference and asymptotic statistics).

- Good computational skills in R or Python (ability to implement algorithms, run simulations, and handle real datasets).

References:

Robinson, P. M. (1988). Root-N-consistent semiparametric regression. *Econometrica*, 931954.

Zhang, C. H. and Zhang, S. S. (2014). Confidence intervals for low dimensional parameters in high dimensional linear models. *JRSSB*, 76(1), 217-242.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W. and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, C1-C68.

Zhang, C. H. and Zhang, S. S. (2014). Confidence intervals for low dimensional parameters in high dimensional linear models. *JRSSB*, 76(1), 217-242.
Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W. and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, C1-C68.

Generating Network with Pre-determined Counts on Edges, Two-star Edges and Triangles

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An applied analysis in network science typically starts with reporting various numerical summaries of network structure. Unfortunately there are typically no algorithms to generate a network matching the three simple and important counts, i.e. the counts on edges, twostar edges and triangles, while leaving all other characteristics of the graph completely random.

To fill in the gap, we have developed an algorithm based on the rewiring idea of Mahadevan et al. (2006). This project is to investigate further on this problem and to improve the algorithm.

References:

a) Chang, J., Kolaczyk, E., and Yao, Q. (2022). Estimation of subgraph densities in noisynetworks. Journal of the American Statistical Association, **117**, 361-374.
b) Mahadevan, P., Krioukov, C., Fall, K. and Vahdat, A. (2006). Systematic topologyanalysis and generation using degree correlations. SIGCOMM'06 Proceedings of the 2006 conference on Applications, technologies, architectures, and protocols for computer communications, 135-146.

Testing for Change-points in Dynamic Networks

Supervisor: Professor Qiwei Yao, Q.Yao@lse.ac.uk

Jiang et al (2023) a simple order-one autoregressive setup for building dynamic network processes. It facilitates effective statistical inference for various network structures such as dynamic stochastic block (DSB) models. Estimation for a change-point in a DSB process has also been investigated. This project is to consider the statistical tests for change-points in a DSB process. More specifically it will investigate the following two aspects.

a) The effectiveness of the permutation test proposed in Jiang et al (2021) for detecting change-points.
b) The likelihood ratio test for the existence of one change-point.

References:

Jiang, B., Li, J. and Yao, Q. (2023). Autoregressive networks. Journal of Machine Learning Research, 24(227), 1-69.

Note. Both projects involve intensive programming and numerical simulation.